CHOWK. Traffic signals that learned lane-free Indian traffic

A learned signal controller and a live 3D simulation in the browser, with a benchmark any visitor can re-run. INNOVEX 2026 · PS04 Smart Cities & Sustainability.

Live demo: https://chowk-seven.vercel.app/ · **Code:** https://github.com/JunaidAhamed-7777/chowk

RESULT (SIMULATION)

> On 20 held-out seeds in simulated Indian mixed traffic, CHOWK reduced delay per person by 4.6% versus area max-pressure (95% CI −7.5% to −1.7%), with 60.0% lower delay versus Webster fixed-time (95% CI −62.7% to −57.2%). Simulation result.

Morning peak, 20 held-out seeds	Delay per person (s)	Stops per vehicle	Throughput (people/h)
CHOWK	**84.0** [75.1, 93.3]	**2.40** [2.30, 2.51]	**6282** [6034, 6517]
Area max-pressure	88.0 [79.2, 97.3]	2.43 [2.33, 2.53]	6285 [6028, 6518]
Count max-pressure	99.8 [91.8, 107.8]	2.15 [2.04, 2.27]	6244 [5982, 6484]
Actuated	125.1 [113.4, 137.4]	2.94 [2.82, 3.04]	6092 [5850, 6316]
Fixed-time (Webster)	209.7 [192.7, 226.1]	3.21 [3.10, 3.31]	5571 [5336, 5787]

CHOWK had the lowest mean delay per person in 6 of 7 scenarios. Means with 95% bootstrap intervals; every number is regenerated from the published runs.

THE PROBLEM

On Indian roads two-wheelers filter through gaps and buses share road width with autos, so counting vehicles misreads a queue, and a bus with forty people counts the same as a car with one.

THE APPROACH

- **Lane-free simulator** in TypeScript: one deterministic package for training, benchmark and the live demo.
- **Area perception, person objective:** the controller sees occupied road area and minimises delay to people.
- **Residual on max-pressure:** a small network learns a correction to area max-pressure, trained with evolution strategies; checkpoints chosen on validation seeds only.

PROOF YOU CAN CHECK

- Pre-registered primary endpoint, held-out seeds never used for training or tuning.
- The Results page replays published episodes in your browser and compares every metric and final state hash; Node, Chromium, WebKit, Firefox reach the same state hash.
- Any comparison can be checked seed by seed; ablations, generalisation and robustness suites are published.

HONEST LIMITS

- All results are simulation results on an authored corridor, not field data.
- The turning model is conservative; driving parameters are documented assumptions.
- A field trial would need local calibration, shadow mode and an authority-approved pilot.

Prior work: Nandakumar, Banerjee and Vanajakshi, arXiv 2509.21745; C-DAC CoSiCoSt. Method and limits: https://chowk-seven.vercel.app/method · https://chowk-seven.vercel.app/limits

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